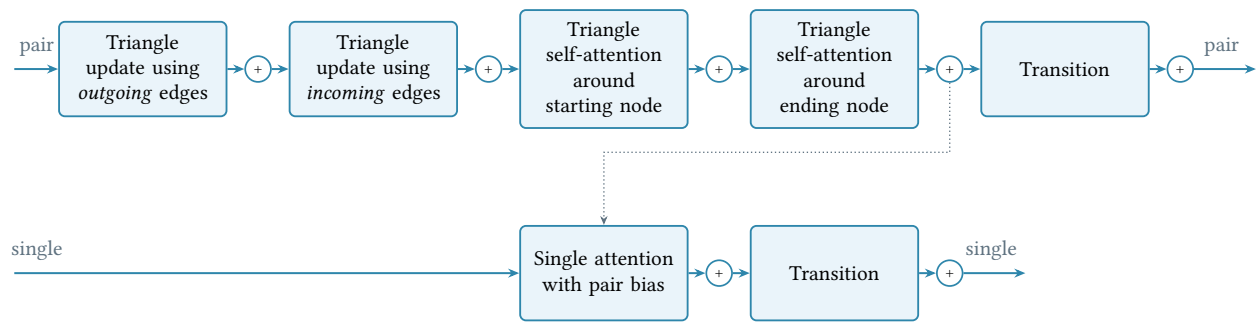


MiniWorld 0

Architecture schematics

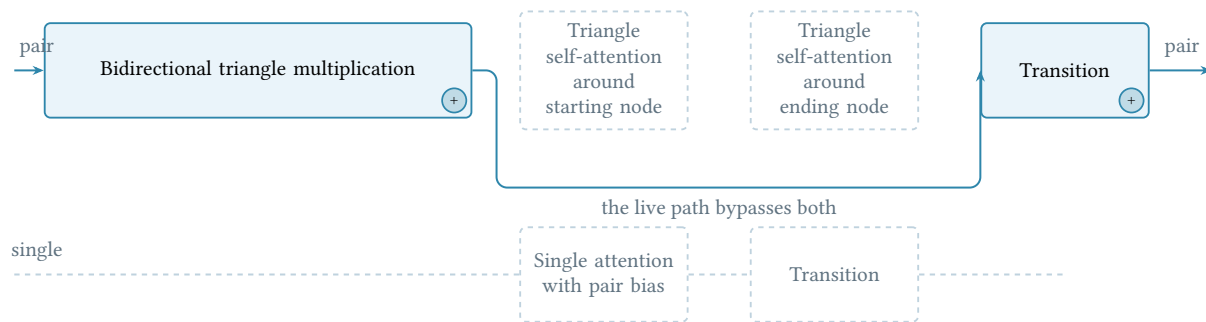
1 The pair update

a AlphaFold 3 pairformer block, $\times 48$



7 sub-blocks · 7 residual additions

b MiniWorld 0 pairformer block, $\times 48$



2 sub-blocks · no external residual additions

 retained removed + residual added by the block + residual fused into the kernel